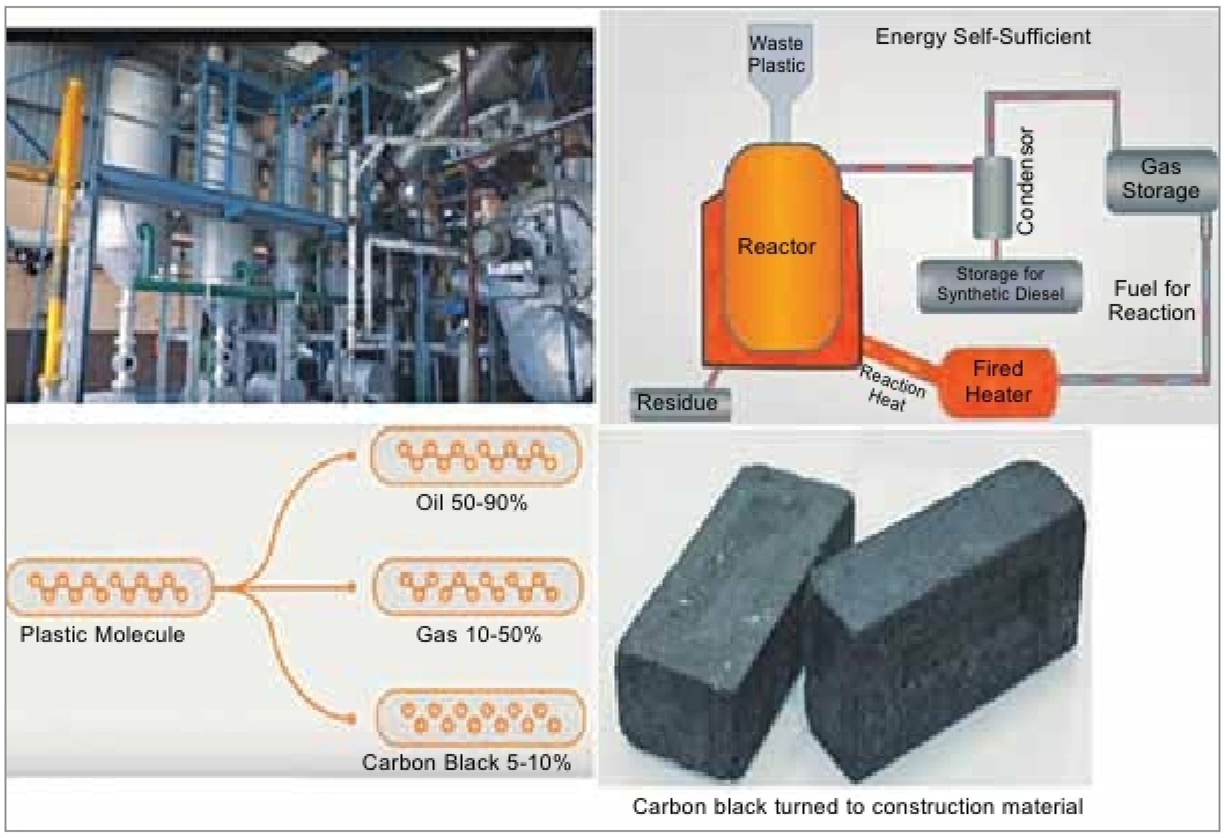

Fig. 2: APChemi plant for recycling via pyrolysis

ability, unfortunately, in India, many WTE plants are being shut down due to environmental violations.

Some reasons behind this failure on our part are:

1. Our WTE plants do not segregate the solid wastes, so wet wastes get mixed with dry wastes and reduce the calorific value of the wastes.
2. The high organic content in the wastes is another factor that decreases their calorific value.
3. Some wastes are highly inert, and the high moisture content in the wastes decreases the quality of it. Hence, incineration plants are unable to burn the wastes completely. In fact, most of the wastes are unfit for incineration.
4. Unsegregated wastes require an excessive amount of fuel to burn. So, the cost of energy per unit from our WTE plants is high. Like the USA and European landfills, Indian landfills should learn to accept only non-recyclable and non-biodegradable wastes, and incineration should be done by grate combustion technology.

Successful Indian companies that turn end-of-life plastics to fuels

When WTE plants have shown their downsides, waste plastics have come as the light at the end of the tunnel, enabling us to move forward and transform end-of-life plastics to renewable fuels by adopting simple and cost-effective ways. The growing interest in creating something different from plastic wastes has led to many methods that turn plastics to fuel(s). Plastics are considered to be toxic, but by following proper waste management techniques, these can be vaporised rather than burnt and oxidised. A brief account of some companies that are successfully turning waste plastics to sustainable fuel(s) is given below.

Indian Institute of Petroleum (IIP), Dehradun, under the umbrella of Council of Scientific & Industrial Research (CSIR), has set up a plant that is the first by a government research institute in India. It converts waste plastics into useful hydrocarbons like petroleum or diesel using pyrolysis method. In this method, the segregated plastic wastes are not burnt but heated in the absence of air, so that the complex hydrocarbons are broken to small molecules. A right combination of catalysts is added to these small molecules, which are then converted to valuable hydrocarbons in vapour form. This vapour is then condensed to get the final products such as gasoline, diesel, or aromatics, in liquid form.

The IIP plant is capable of converting one tonne of plastic wastes to 850 litres of diesel, or 700 litres of petrol, or 500 litres of aromatic products. The obtained fuels are free from toxic substances and are thus completely environment friendly. The final IIP products are welcome by Indian Railways because these have zero sulphur and therefore meet the Euro-III standards. The aromatics are rich in BTX.

Agile Process Chemicals (APChemi) is a leading solution provider in following the technology of chemical recycling via pyrolysis. The company is one among the eleven winners of the Asia-Pacific End Plastic Waste Innovation Platform, which supports the innovations of the startups in ending waste plastics, conducted by Alliance to End Plastic Waste & Plug and Play.

At APChemi, the plastics from the municipal solid wastes are thoroughly washed, dried, shredded and then fed into a gas-powered reactor where the plastic, which contains complex hydrocarbons, is broken down using heat to its molecular level. It yields oil and gas that are clean and efficient hydrocarbons, which serve as fuels and petrochemicals, thus reducing the dependence on crude oil and paving the way to an ecological circular economy.

The only waste product in the pyrolysis process is the carbon black. However, APChemi has successfully converted this pyrolysis waste to solid sustainable construction material. APChemi's plant successfully converts 75,000kg of end-of-life plastics per day,

Fig. 3: Plant of 500kg capacity at Rudra Environmental Solutions